EE1101: Circuits and Network Analysis

Assignment - 13

Handed out: 09 - Nov - 2024 **Due : 28 - Nov - 2024 (before 5 PM)**

Instructions :

1. Please upload your assignment solutions to the course page on the Canvas platform. Only solutions submitted through canvas will be reviewed. For specific guidelines, refer to the instructions provided on the course page.
2. It is suggested that you attempt all the problems. However, it is sufficient to submit solutions for problems that total 10 points.
3. Submissions received after the deadline will attract negative marking. Ensure that your submissions are named in the following format: RollNo-Assignment-13.pdf.

1. (8 points) Consider the circuit shown in Fig. 1.

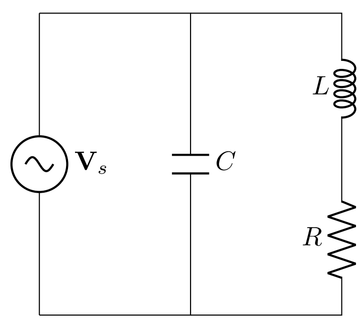

Fig. 1: Circuit for question 1

   (a) (2 points) Determine the frequency at which the current from the source reaches the maximum amplitude.

   (b) (2 points) Compute the energy stored in the circuit when the circuit is operating at resonance. Assume $R = 1\Omega$, $L = 0.1$H and $C = 0.01$F.

   (c) (2 points) Compute the quality factor of the circuit and the bandwidth when $R = 1\Omega$, $L = 0.1$H and $C = 0.01$F.

   (d) (2 points) Compute the active and reactive power delivered by the source at resonance. Assume $R = 1\Omega$, $L = 0.1$H, $C = 0.01$F and $\mathbf{V}_s = 100\angle 0$.

2. (8 points) Consider the circuit shown in Fig. 2.

   (a) (2 points) Determine the frequency at which the current from the source reaches the maximum amplitude.

   (b) (2 points) Compute the energy stored in the circuit when the circuit is operating at resonance. Assume $R = 1\Omega$, $L = 0.1$H and $C = 0.01$F.

   (c) (2 points) Compute the quality factor of the circuit and the bandwidth when $R = 1\Omega$, $L = 0.1$H and $C = 0.01$F.

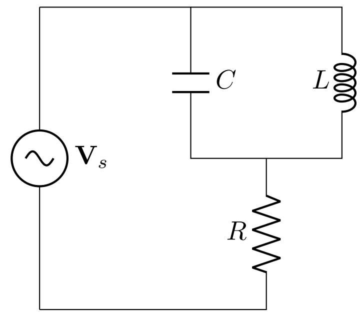

Fig. 2: Circuit for question 2

(d) (2 points) Compute the active and reactive power delivered by the source at resonance. Assume $R = 1\Omega$, $L = 0.1$H, $C = 0.01$F and $\mathbf{V}_s = 100\angle 0$.

3. (2 points) Derive an expression for resonant frequency for the circuit shown in Fig. 3.

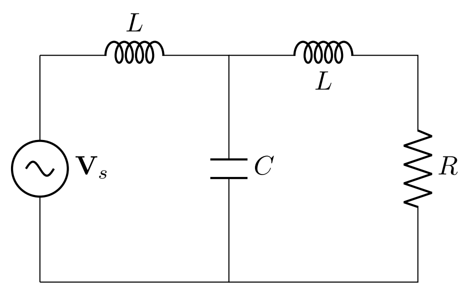

Fig. 3: Circuit for question 3

4. (4 points) It is desired to design a RC low-pass filter whose bandwidth is 10 kHz. Determine the value of C if R is chosen to be 1kΩ. Plot the frequency response of the designed low pass filter.

5. (4 points) Consider a scenario where two low-pass RC filters are cascaded together. For the purpose of analysis, consider the RC filter designed in question 4. Obtain the frequency response of the cascaded system. Also comment on the bandwidth of the cascaded system.